Systems of Equations Maze

Name _____ Period ____

Start in the top-left room. Solve the system and write the solution as a point (x, y) or as No solution / Infinitely many, then go through the door with your answer.

START

$y = 2x$
$x + y = 9$

(3, 6)

$y = x + 1$
$y = -x + 5$

(1, 4)

$x + 2y = 9$
$x - y = 0$

(0, −7)

$3x - y = 7$
$x + y = -3$

(−3, −6)

(6, 3)

(2, 3)

(3, 2)

(3, −3)

(−3, −3)

(−4, 1)

$2x + 3y = 7$
$x - 3y = -1$

(1, 2)

$x + y = 10$
$x - y = 4$

(3, 7)

$y = -x + 4$
$y = 3x - 8$

(3, −1)

$y = 4x + 1$
$2x + y = -5$

(2, 1)

(3, 5)

(7, 3)

(1, 3)

(−1, 3)

(3, −4)

(−4, −3)

$4x - 2y = 6$
$y = 2x + 1$

(0, 1)

$y = 3x - 4$
$2x + y = 11$

(−3, 5)

$6x - 3y = 9$
$y = 2x - 3$

Infinitely many

$5x + 2y = -14$
$3x - 4y = -24$

No solution

(5, 3)

(1, −1)

(7, −3)

(1, 1)

(−2, −2)

(−4, 3)

$3x + 2y = 4$
$x - y = 3$

(2, −1)

$2x + 5y = -1$
$3x - 2y = 27$

(−3, 7)

$2x + y = 1$
$3x - 2y = 12$

(−3, 2)

FINISH

You made it!

Systems of Equations Maze: Answer Grid

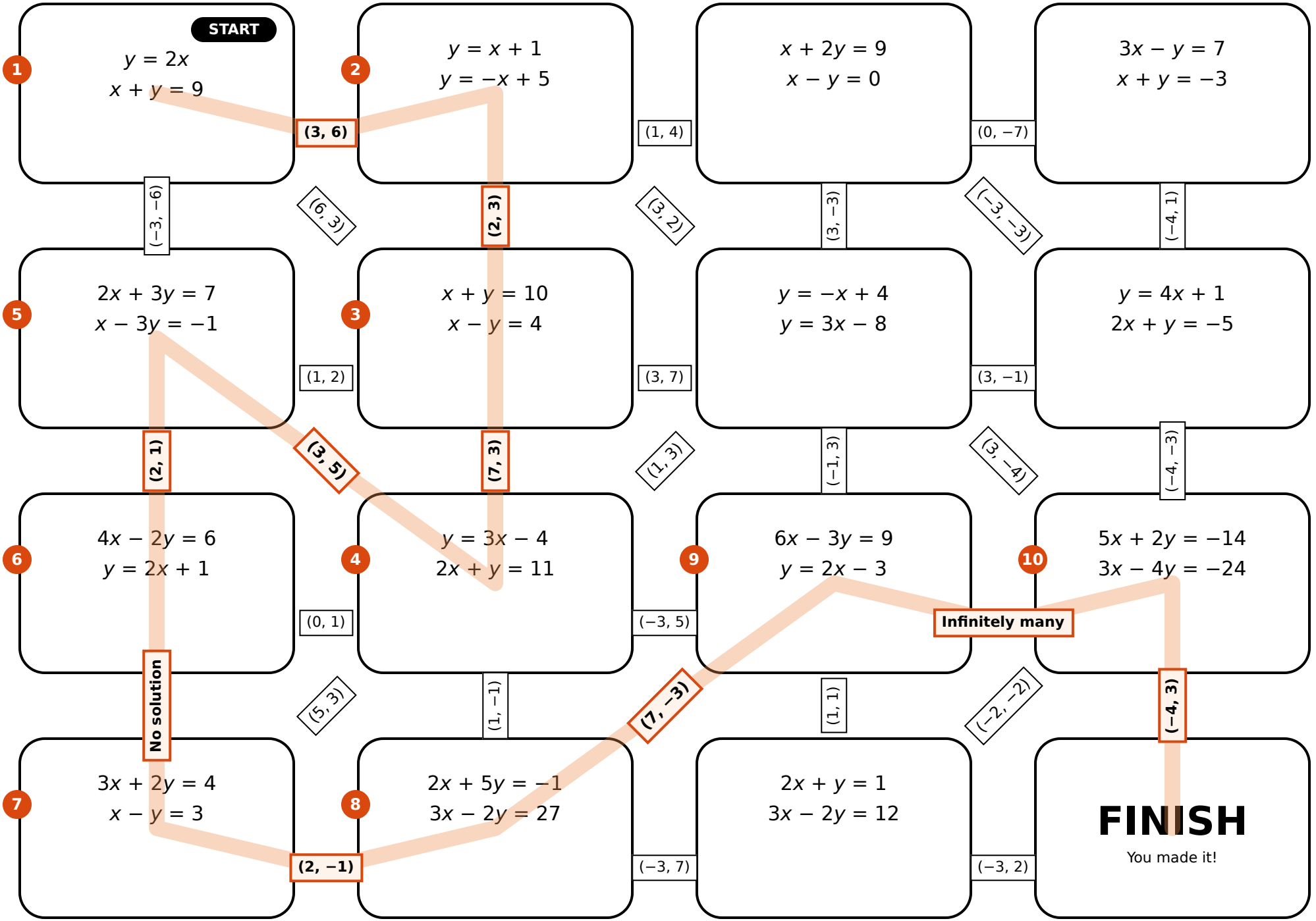
Name _____ Period ____

Write each answer in the order you found it, and show your work for that room.

1	Answer: _____	Work:
2	Answer: _____	Work:
3	Answer: _____	Work:
4	Answer: _____	Work:
5	Answer: _____	Work:
6	Answer: _____	Work:
7	Answer: _____	Work:
8	Answer: _____	Work:
9	Answer: _____	Work:
10	Answer: _____	Work:

Systems of Equations Maze ANSWER KEY

Start in the top-left room. Solve the system and write the solution as a point (x, y) or as No solution / Infinitely many, then go through the door with your answer.



Systems of Equations Maze: Answer Grid **ANSWER KEY**

Write each answer in the order you found it, and show your work for that room.

1	Answer: **(3, 6)**	$x + 2x = 9 \rightarrow x = 3 \rightarrow y = 2(3) = 6$
2	Answer: **(2, 3)**	$x + 1 = -x + 5 \rightarrow 2x = 4, x = 2 \rightarrow y = 2 + 1 = 3$
3	Answer: **(7, 3)**	add: $2x = 14 \rightarrow x = 7 \rightarrow 7 + y = 10, y = 3$
4	Answer: **(3, 5)**	$2x + 3x - 4 = 11 \rightarrow 5x = 15, x = 3 \rightarrow y = 3(3) - 4 = 5$
5	Answer: **(2, 1)**	add: $3x = 6 \rightarrow x = 2 \rightarrow 2(2) + 3y = 7, y = 1$
6	Answer: **No solution**	$4x - 2(2x + 1) = 6 \rightarrow -2 = 6 \rightarrow$ false: parallel lines
7	Answer: **(2, −1)**	$x = y + 3 \rightarrow 3(y + 3) + 2y = 4 \rightarrow 5y = -5, y = -1 \rightarrow x = -1 + 3 = 2$
8	Answer: **(7, −3)**	$4x + 10y = -2$; $15x - 10y = 135 \rightarrow 19x = 133, x = 7 \rightarrow 14 + 5y = -1, y = -3$
9	Answer: **Infinitely many**	$6x - 3(2x - 3) = 9 \rightarrow 9 = 9 \rightarrow$ true: same line
10	Answer: **(−4, 3)**	$10x + 4y = -28$; $3x - 4y = -24 \rightarrow 13x = -52, x = -4 \rightarrow -20 + 2y = -14, y = 3$